THE SECOND DIAGONAL AXIOM PLUS THE AXIOMS IMPLY ALGEBRAICITY

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 2782-2796 of sources/stacks-artin/pinned-original-source.tex; proof span is lines 2798-2851, the complete author proof (52 body lines).

Editorial note. Navigation stubs: 15 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. Queue index 15 of queue_artin.json (pb=2471, stmt 2782-2796, proof 2798-2851).

Editorial note. stacks-artin wave ar01 (2026-10-14): fresh adjudication accept per lane-s3/artin_judgments.md; pool per Chief 10:5x direct dispatch (idle-first over S12's 01:5x note, SMC precedent); artin.tex 99 rows/67 with proof judged 46 accept / 21 reject / 5 AGP-overlap avoidance rows; patch-sd-envs contract spans scanned.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

Source reference. Original proposition 07WH: sources/stacks-artin/pinned-original-source.tex, source lines 2851–2851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07WG: sources/stacks-artin/pinned-original-source.tex, source lines 2851–2851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07WQ: sources/stacks-artin/pinned-original-source.tex, source lines 743–746. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07Y4: sources/stacks-artin/pinned-original-source.tex, source lines 2729–2741. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07X8: sources/stacks-artin/pinned-original-source.tex, source lines 1323–1326. Cited by the retained statement or proof but

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not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07WR: sources/stacks-artin/pinned-original-source.tex, source lines 753–759. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07X9: sources/stacks-artin/pinned-original-source.tex, source lines 1438–1445. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07XM: sources/stacks-artin/pinned-original-source.tex, source lines 1693–1706. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07X2: sources/stacks-artin/pinned-original-source.tex, source lines 1154–1178. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07X1: sources/stacks-artin/pinned-original-source.tex, source lines 1030–1037. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07Y1: sources/stacks-artin/pinned-original-source.tex, source lines 2641–2649. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 07XJ: sources/stacks-artin/pinned-original-source.tex, source lines 2462–2489. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 05UI: sources/stacks-artin/pinned-original-source.tex, source lines 2851–2851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 04SI: sources/stacks-artin/pinned-original-source.tex, source lines 2851–2851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original proposition 0432: sources/stacks-artin/pinned-original-source.tex, source lines 2851–2851. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Proposition 0.1. *Let S be a locally Noetherian scheme. Let $p : \mathcal{X} \to (Sch/S)_{fppf}$ be a category fibred in groupoids. Assume that*

1. *$\Delta_\Delta : \mathcal{X} \to \mathcal{X} \times_{\mathcal{X}\times\mathcal{X}} \mathcal{X}$ is representable by algebraic spaces,*
2. *$\mathcal{X}$ satisfies axioms [-1], [0], [1], [2], [3], [4], and [5] (see Section 07XJ),*
3. *$\mathcal{O}_{S,s}$ is a G-ring for all finite type points s of S.*

Then $\mathcal{X}$ is an algebraic stack.

Proof. We first prove that $\Delta : \mathcal{X} \to \mathcal{X} \times \mathcal{X}$ is representable by algebraic spaces. To do this it suffices to show that

$$\mathcal{Y} = \mathcal{X} \times_{\Delta, \mathcal{X} \times \mathcal{X}, y} (Sch/V)_{fppf}$$

is representable by an algebraic space for any affine scheme V locally of finite presentation over S and object y of $\mathcal{X} \times \mathcal{X}$ over V, see Criteria for Representability, Lemma 07WH[1]. Observe that $\mathcal{Y}$ is fibred in setoids (Stacks, Lemma 04SI) and let $Y : (Sch/S)^{opp}_{fppf} \to Sets$, $T \mapsto \mathrm{Ob}(\mathcal{Y}_T)/\cong$ be the functor of isomorphism classes. We will apply Proposition 07Y1 to see that Y is an algebraic space.

Note that $\Delta_{\mathcal{Y}} : \mathcal{Y} \to \mathcal{Y} \times \mathcal{Y}$ (and hence also $Y \to Y \times Y$) is representable by algebraic spaces by condition (1) and Criteria for Representability, Lemma 07WG. Observe that Y is a sheaf for the étale topology by Stacks, Lemmas 0432 and 05UI, i.e., axiom [0] holds. Also Y is limit preserving by Lemma 07XM, i.e., we have [1]. Note that Y has (RS), i.e., axiom [2] holds, by Lemmas 07WQ and 07WR. Axiom [3] for Y follows from Lemmas 07X1 and 07X2. Axiom [4] follows from Lemmas 07X8 and 07X9. Axiom [5] for Y follows directly from openness of versality for $\Delta_{\mathcal{X}}$ which is part of axiom [5] for $\mathcal{X}$. Thus all the assumptions of Proposition 07Y1 are satisfied and Y is an algebraic space.

At this point it follows from Lemma 07Y4 that $\mathcal{X}$ is an algebraic stack. □

[1]The set theoretic condition in Criteria for Representability, Lemma 07WH will hold: the size of the algebraic space Y representing $\mathcal{Y}$ is suitably bounded. Namely, $Y \to S$ will be locally of finite type and Y will satisfy axiom [-1]. Details omitted.